

RAUTAVAARA YLA-LUOSTA, Finland

WMO: 027890

Lat: 63.3781N

Lon: 28.6617E

Elev: 166

StdP: 99.35

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.6	(c) -26.5	(d) -32.7	(e) 0.2	(f) -29.6	(g) -29.2	(h) 0.3	(i) -26.4	(j) 7.0	(k) -7.7	(l) 6.1	(m) -5.6	(n) 0.8	(o) 0	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 26.8	(d) 18.2	(e) 24.7	(f) 17.2	(g) 22.7	(h) 16.2	(i) 19.7	(j) 24.5	(k) 18.5	(l) 23.0	(m) 17.3	(n) 21.2	(o) 2.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.1	21.5	16.8	12.2	20.5	15.6	11.3	19.2	56.8	24.3	53.0	23.5	49.3	21.1	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.8	(b) 5.2	(c) 4.6	(e) -32.9	(f) 28.7	(g) 3.2	(h) 3.0	(i) -35.2	(j) 30.8	(k) -37.1	(l) 32.6	(m) -38.9	(n) 34.3	(o) -41.2	(p) 36.5		
(a) 5.8	(b) 5.2	(c) 4.6	(e) -32.8	(f) 21.1	(g) 3.1	(h) 1.4	(i) -35.0	(j) 22.1	(k) -36.9	(l) 23.0	(m) -38.6	(n) 23.7	(o) -40.9	(p) 24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-11.8	-8.7	-5.1	1.3	8.5	12.5	15.9	14.0	9.2	2.4	-1.6	-5.8	(6)
(7)		DBStd	10.34	8.24	7.19	5.52	3.99	4.63	4.02	3.66	3.32	3.37	3.63	4.56	6.56	(7)
(8)		HDD10.0	3169	674	525	469	262	83	19	2	6	54	237	347	491	(8)
(9)		HDD18.3	5766	933	758	727	512	305	181	92	141	275	495	597	749	(9)
(10)		CDD10.0	474	0	0	0	0	37	93	186	128	29	1	0	0	(10)
(11)		CDD18.3	28	0	0	0	0	1	4	18	5	0	0	0	0	(11)
(12)		CDH23.3	329	0	0	0	0	29	63	177	60	0	0	0	0	(12)
(13)		CDH26.7	59	0	0	0	0	2	9	38	11	0	0	0	0	(13)
(14)	Wind	WSAvg	2.2	1.8	2.3	2.5	2.3	2.4	2.3	2.0	2.1	2.1	2.2	2.3	2.2	(14)
(15)	Precipitation	PrecAvg	669	48	41	36	33	49	70	87	72	59	59	57	58	(15)
(16)		PrecMax	875	86	87	71	56	106	138	163	158	113	112	125	114	(16)
(17)		PrecMin	512	14	5	7	8	6	23	24	16	15	12	9	19	(17)
(18)		PrecStd	90	18	19	15	14	24	26	36	31	26	26	25	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	4.1	7.7	16.8	26.4	28.0	29.6	28.2	20.2	11.7	8.1	4.1	(19)
(20)			MCWB	2.0	2.4	3.2	9.3	14.4	17.5	20.2	19.6	15.5	10.3	7.6	3.6	(20)
(21)		2%	DB	0.7	2.0	4.8	12.7	23.5	25.3	27.4	24.8	17.8	10.3	6.0	2.7	(21)
(22)			MCWB	0.4	1.3	1.6	7.3	13.9	17.6	19.0	17.7	13.8	9.2	5.6	2.4	(22)
(23)		5%	DB	0.0	0.7	3.0	10.4	20.6	22.7	25.2	22.4	16.1	9.0	4.7	1.7	(23)
(24)			MCWB	-0.2	0.2	0.9	5.4	12.7	15.5	18.2	16.5	13.3	8.0	4.4	1.3	(24)
(25)		10%	DB	-0.9	0.1	1.7	8.0	17.4	20.5	23.1	20.5	14.7	7.7	3.5	0.7	(25)
(26)			MCWB	-1.1	-0.2	0.2	4.1	10.8	14.3	17.2	15.9	12.7	6.9	3.3	0.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.5	3.9	10.0	16.2	19.7	21.5	20.9	16.1	11.1	7.7	3.9	(27)
(28)			MCDWB	2.3	3.3	6.7	15.0	22.9	24.1	27.2	26.9	19.4	11.6	8.1	4.1	(28)
(29)		2%	WB	0.5	1.4	2.4	7.8	14.7	18.3	20.2	18.5	14.9	9.6	5.7	2.5	(29)
(30)			MCDWB	0.6	2.0	3.9	12.7	21.3	23.6	25.2	23.0	16.6	10.1	5.9	2.7	(30)
(31)		5%	WB	-0.1	0.5	1.4	5.8	13.3	16.3	19.1	17.5	13.9	8.2	4.4	1.3	(31)
(32)			MCDWB	0.0	0.9	2.5	9.2	19.3	21.4	23.9	21.1	15.8	8.8	4.7	1.5	(32)
(33)		10%	WB	-1.1	-0.2	0.6	4.5	11.4	15.0	18.1	16.5	12.9	6.9	3.3	0.5	(33)
(34)			MCDWB	-0.9	0.1	1.4	7.2	16.7	19.2	22.1	19.7	14.3	7.6	3.5	0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	7.0	9.3	9.7	10.7	10.1	9.6	9.6	7.6	5.2	3.9	4.8	(35)
(36)			MCDBR	3.3	4.8	9.1	14.4	16.0	13.7	12.4	13.2	10.6	7.1	4.3	3.3	(36)
(37)			MCWBR	3.5	4.5	6.6	8.7	7.5	6.6	5.4	6.7	7.2	5.9	4.3	3.2	(37)
(38)		5% WB	MCDBR	3.3	4.7	7.1	12.1	14.3	11.9	11.4	10.8	9.4	6.6	4.3	3.3	(38)
(39)			MCWBR	3.4	4.4	5.5	7.6	7.3	6.1	5.6	6.0	6.9	5.9	4.3	3.3	(39)
(40)			Clear-Sky Solar Irradiance	taub	0.221	0.280	0.308	0.346	0.346	0.343	0.368	0.350	0.327	0.307	0.236	0.208
(41)	taud	1.987		2.110	2.171	2.288	2.453	2.485	2.430	2.467	2.519	2.399	2.074	1.968	(41)	
(42)	Ebn at Noon	381		635	778	832	870	879	841	826	777	627	359	185	(42)	
(43)	Edh at Noon	31		65	92	103	96	95	98	87	68	51	28	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.18	0.73	2.21	3.67	4.72	5.25	4.97	3.78	2.07	0.78	0.19	0.07	(44)	
(45)		RadStd	0.02	0.17	0.26	0.42	0.52	0.33	0.52	0.42	0.24	0.13	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	-177	-191	N/A	N/A

Nomenclature: See separate page